Pro Football Sports Science Research Update

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Change in head impact exposure following return to sport in concussed football athletes. Medicine and Science in Sports and Exercise, Ahead of Print: 1-45, 2025.

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Detection of injury biomarkers in sweat of collegiate athletes pre- and post-football season: A pilot study. Neurotrauma Reports, Vol 6: 1-6, 2025.

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Effects of sleep deprivation on sports performance and perceived exertion in athletes and non-athletes: A systematic review and meta-analysis. Frontiers in Physiology, 16: 1544286, 2025.

Evening smartphone exposure impairs sleep quality and next-day performance in elite soccer players: A randomized controlled trial. Biology of Sport, 43: 227-242, 2026.

Hamstring injury mechanisms and eccentric training-induced muscle adaptations: Current insights and future directions. Sports Medicine, Ahead of Print: 1-15, 2025.

Heavy hitters, light sleepers: Collision frequency and locomotor load on sleep architecture in professional Rugby Union players. European Journal of Sport Science, 25: e70052, 2025.

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Tackle characteristics associated with concussion in elite men's rugby union: Unpicking the differences between tacklers and ball-carriers. BMJ Open Sport & Exercise Medicine, 11: e002612, 2025.

Using a Bayesian network to classify time to return to sport based on football injury epidemiological data. PLOS ONE, 20(3): e0314184, 2025.

Acute fatigue-induced alterations in hamstring muscle properties after repeated Nordic hamstring exercises. Science Progress, 107(2): 1-17, 2024.

Abstract - This study aimed to investigate the impact of the Nordic hamstring exercises (NHE) on acute fatigue-induced alterations in the mechanical and morphological properties of hamstring muscles. The second aim was to define the blood flow and perfusion after NHE in recreational active volunteers. Twenty-two individuals volunteered to participate in the study. This study investigated fatigue outcomes: rate of perceived exertion (RPE) scale and average force generated during NHE; mechanical properties (stiffness); morphological properties (thickness, pennation angle, and fascicle length), and vascularity index (VI) of the semitendinosus (ST) and biceps femoris long head (BFLH) at baseline, immediately post-exercise and 1-h post-exercise. The NHE fatigue procedure consisted of six bouts of five repetitions. The results showed an increase in thickness and pennation angle of BFLH and ST immediately post-exercise and a decrease in thickness and pennation angle of BFLH and ST 1-h post-exercise. While the fascicle length of BFLH and ST decreased immediately post-exercise and increased 1-h post-exercise. The VI for two muscles increased immediately post-exercise and after 1-h post-exercise. Moreover, we found a relationship between RPE and average force, that is, as RPE increased during NHE, average force decreased. In conclusion, eccentric NHE exercises significantly and acutely affect BFLH and ST. The NHE fatigue protocol significantly affected the mechanical and morphological properties of BFLH and ST muscles, changing their thickness, fascicle length, pennation angle, and VI.

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Annual Banned-Substance Review 17th Edition – Analytical approaches in human sports drug testing 2023-2024. Drug Testing and Analysis, 17: 1417-1442, 2025.

Abstract - The 17th edition of the annual banned-substance review on analytical approaches in human sports drug testing is dedicated to literature published between October 2023 and September 2024. As in previous years, focus is put particularly on new or enhanced analytical options in human doping controls as well as investigations into the metabolism and elimination of compounds of interest, which represent central (while not exclusive) cornerstones of the global anti-doping mission. New information published within the past 12 months on established doping agents as well as new potentially relevant substances are reviewed and discussed in the context of the World Anti-Doping Agency's 2024 Prohibited List. Thereby, analytical challenges, especially with regard to the continuously growing number of target compounds and potentially relevant drug classes as well as the exigency (and consequences) of utmost analytical retrospectivity, are thematized and contextualized. Investigations especially into anabolic agents, peptide hormones, and strategies for the detection of gene doping were identified as core areas of anti-doping research in the reviewed period.

Biomechanical determinants of change of direction performance: A systematic review. Sports Medicine, Ahead of Print: 1-18, 2025.

Background: The ability to change direction rapidly is crucial for enhancing performance in multidirectional sports. Evidence suggests that several biomechanical variables are associated with faster change of direction (COD) completion times. However, while it is understood that biomechanical factors influence COD performance, the evidence remains unclear because of the diverse range of biomechanical factors, inconsistent findings and potential influences from various moderating factors (e.g. sex, training experience).

Objective: The primary aim of this systematic review was to identify the biomechanical determinants of COD performance while the secondary aim was to examine the impact of moderating factors on the determinants. The findings of this review could assist practitioners in designing effective training and coaching strategies to improve COD performance.

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Methods: A systematic literature search was conducted across the electronic databases of Scopus, PubMed, Web of Science, CINAHL, and SPORTDiscus. Studies were considered eligible if they involved healthy participants, considered biomechanical determinants of COD performance via correlational analyses and reported COD performance (i.e. time to completion). The quality of the study was assessed via the Kmet scale while study findings were collated.

Results: A total of 13 studies met the inclusion criteria and analyzed 45-180° COD tasks involving 374 participants. Kmet scores ranged from 73 to 96%, indicating good-to-excellent methodological quality of studies. Several biomechanical variables were identified as contributors to quicker COD completion times, including shorter ground contact time, higher approach and exit velocities, increased braking and propulsive forces, greater trunk inclination angle, lower center-of-mass height, and increased moments and power at the hip, knee and ankle. With respect to moderating factors, included studies utilized various COD tasks (45-180°), examined mostly male participants (79.4%) with inconsistent reporting of playing/training experience and all consisted of a pre-planned COD task only.

Conclusions: Our findings identified several key biomechanical variables that were important determinants of faster COD performance. However, the impact of moderating factors on COD performance was minimally examined in prior studies and requires further investigation. Recommendations are provided in this paper focusing on biomechanical contributors (e.g. ground contact time, approach velocity, braking forces), which may assist coaches with relevant training modalities to enhance COD performance.

Caffeine and Sports Performance: The conflict between caffeine intake to enhance performance and avoiding caffeine to ensure sleep quality. Sports Medicine, 55: 1579-1592, 2025.

Abstract - The ergogenic effects of caffeine have been reported in scientific literature over a wide spectrum of sporting activities. The current recommendation for caffeine supplementation is ingesting ~ 3-6 mg/kg about 1 h before the onset of exercise. However, some studies reporting caffeine-induced ergogenic effects during exercise have also reported increased activation and reduced sleep quality in the hours after caffeine ingestion. While most of the research on caffeine supplementation for sporting

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activities recommends the consumption of this stimulant to enhance performance, research focusing on athletes' sleep quality advises against this decision, especially before competition in the evening or later. Considering that some athletes often compete in the evening or later, the general recommendation of caffeine supplementation may be modified for these athletes as acute caffeine intake in these conditions may produce undesirable side effects such as insomnia and, potentially, reduced performance in subsequent days. In this review, we examine literature on this topic to help athletes and sports practitioners solve the dilemma between the convenience of using caffeine to enhance sports performance or avoiding caffeine to ensure sleep quality. This review identifies potential solutions for this decision, keeping the focus on athletes' well-being. Overall, the performance response to caffeine and the effect of this substance on sleep quality can vary inter-individually and depend on the conditions of the exercise session (time of onset, duration, etc.). For this reason, nutritional practitioners should assess their athletes individually to resolve the conflict between caffeine ingestion and ensuring sleep quality on an individual basis, using simulated competitions with dual measurement of performance during exercise and unwanted effects in the following hours. In addition, the dose, timing, and source of caffeine supplementation can be individually adjusted to obtain performance benefits while reducing side effects for athletes ingesting caffeine before evening sporting events.

Calf muscle injury in elite athletics: An 8-season prospective cohort study. Journal of Science and Medicine in Sport, Ahead of Print: 1-23, 2025.

Objectives: To describe epidemiological data of indirect calf muscle injury (CMI) in elite track and field athletes. To investigate relationships between CMI and athlete characteristics to inform future injury prevention strategies.

Design: Descriptive epidemiology study.

Methods: A total of 201 track and field athletes, mean age 25.9 ± 4.4, were prospectively observed over 8 seasons (616 athlete seasons). The main outcome measures were injury location, injury number, incidence, incidence proportion, severity, burden and recurrence. Poisson generalised that linear regression models were used to explore relationships between potential risk factors.

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Results: There were 85 CMIs during the study period. Thirty three percent of CMIs were classified as severe. Soleus injuries had the highest incidence, burden and recurrence and were present across all event groups. Incidence proportion of all CMIs was 11.1 %, with the highest incidence of injuries occurring in April. Acute CMI was the most common mechanism, with the right lower limb (58 %) being more affected than the left (42 %). There was no association between CMI occurrence and sex, though there was a significant positive effect between both age and previous lower limb injury.

Conclusions: Calf muscle injury is prevalent in elite track and field athletes. CMI has a peak incidence in April which can have a detrimental effect on an athlete's performance goals for the impending summer competition season. The soleus muscle is the most injured muscle with the biggest burden and recurrence making it a muscle of focus when developing rehabilitation and prevention programs. Older athletes, and athletes with a previous lower limb injury, are more at risk of CMI in elite athletics. Due to the burden and potential performance impact of CMI, further injury prevention efforts are required to reduce the impact of this injury type.

Change in head impact exposure following return to sport in concussed football athletes. Medicine and Science in Sports and Exercise, Ahead of Print: 1-45, 2025.

Purpose: Following sport-related concussions, early head impact exposure and premature return-to-sport are known to increase risk of repeat concussion in football athletes, yet athletes' true post-injury head impact exposure profiles (i.e., characteristics of recorded head impacts over a given time period) and biomechanical progression have not been explored. Accordingly, this study explored how head impact exposure in American college football athletes was altered during their return to sport from concussion, particularly within the same athletic season.

Methods: This analysis compared daily volume of head impacts following concussion with pre-injury levels using head impact exposure profiles of fifty-two concussed collegiate football athletes from six NCAA Division I programs, and further compared these athletes to team- and position-matched controls to minimize season- or team-related factors. Additionally, this study provided an analysis of the possible

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association between duration of recovery and change in head impact exposure following concussion using continuous linear regression.

Results: When comparing to pre-injury levels, 75% of concussed athletes reduced their head impact exposure in their immediate return-to-sport, while over 40% of concussed athletes did not reach their pre-injury level of head impact exposure at any point during the remainder of the concussion season segment. Furthermore, concussed athletes significantly decreased their head impact exposure over their immediate return-to-sport period when compared to team- and position-matched healthy, non-concussed athletes over the same time period. Finally, longer post-concussion recovery times were associated with larger decreases in head impact exposure after return-to-sport.

Conclusions: This study provides evidence for a shift in head impact exposure after returning from concussion, seen most strongly in the immediate days after return-to-sport. These findings align with the recent shift toward more conservative post-concussion management seen across multiple sports and playing levels.

Closed kinetic chain upper extremity stability test reference values for NCAA American football. International Journal of Sports Physical Therapy, 20(9): 1338-1344, 2025.

Background: The Closed Kinetic Chain Upper Extremity Stability Test (CKCUEST) is a valid and reliable test commonly used during return to sport testing. While reference values have been determined for multiple sports, few have specifically considered National Collegiate Athletic Association (NCAA) American football players.

Purpose: To provide reference values on the CKCUEST for NCAA football players with considerations for position, history of upper extremity injury, year in school, and division of college football.

Study design: Cross-sectional study.

Methods: Eighty football players from three NCAA schools were recruited to participate in this study. Height, mass, position, year in school, and history of upper extremity injury were collected. Subjects

completed three trials of the CKCUEST with their means recorded. Means were compared using ANOVA with post hoc Bonferroni tests. A T-test was used for a secondary comparison of mean touches of linemen versus all other positions. Pearson correlation coefficients assessed the relationship between number of touches and subject characteristics.

Results: Subjects were all males with a mean of 2.5 years in school (95% CI 2.18-2.85). Subjects of the study averaged 27.3 touches. There were no statistically significant performance differences between players from NCAA divisions, with/without injury history, years in school, or athlete demographics. Linebackers had significantly more touches compared to offensive linemen, 29.7 and 24.8 ($p<0.03$), respectively. Additionally, linemen as a group (offensive and defensive) averaged 25.7 touches, which was significantly lower compared to all other positions combined (28.1 touches, $p<0.01$). The minimal detectable change was 1.2 touches.

Conclusion: A reference value of 27 touches was found for the CKCUEST in NCAA football players. A change of at least two touches would indicate a measurable change. Position differences should play a role in clearance for return to sport; specifically, linemen may produce less touches than other non-lineman positions.

Level of evidence: Level 2c: Cross-sectional study.

Daily Use of a Broad-Spectrum Cannabidiol Supplement Produces Detectable Concentrations of Cannabinoids in Urine Prohibited by the World Anti-Doping Agency: An Effect Amplified by Exercise. Medicine and Science in Sports and Exercise, Ahead of Print: 1-41, 2025.

Background: Cannabidiol (CBD), a non-intoxicating phytocannabinoid, is used by athletes to enhance recovery and manage other conditions (e.g., poor sleep, anxiety). Although CBD is not prohibited by the World Anti-Doping Agency (WADA), other cannabinoids found in "broad-spectrum" CBD products (e.g., cannabigerol (CBG), cannabidivarin (CBDV)), remain prohibited.

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Objectives: This study aimed to determine whether 10-weeks' use of a broad-spectrum CBD product (150 mg.day [containing trace concentrations of CBG]) could lead to detectable concentrations of prohibited cannabinoids in urine and plasma. The influence of moderate-intensity exercise was also assessed.

Methods: Thirty-six healthy individuals (47% male) self-administered either a broad-spectrum CBD product (n = 31; CBD) or a visually identical placebo (n = 5; PLA) for 10-weeks. After 10-weeks, participants completed a fasted, 90-minute bout of moderate-intensity exercise (55% VO2peak). Blood and urine samples were collected at baseline (pre-supplementation) and pre- and post-exercise.

Results: No cannabinoids or metabolites were detected at baseline in either the PLA or CBD group. Following 10-weeks of supplementation, urinary concentrations of CBD and its metabolites (6-OH-CBD, 7-COOH-CBD, 7-OH-CBD) were present. CBG and CBDV were also detected in 42% and 68% of pre-exercise samples, respectively. Urinary concentrations of 6-OH-CBD ($P = 0.006$), 7-OH-CBD ($P = 0.009$), CBD ($P = 0.043$), CBG ($P = 0.0023$) and CBDV ($P = 0.033$) also increased from pre- to post-exercise. CBG and CBDV were detected in 74% and 84% of post-exercise samples, respectively. Concentrations of Δ9-THC or its metabolites (11-OH-THC, 11-COOH-THC) were not present at any timepoint.

Conclusions: Daily use of a broad-spectrum CBD supplement resulted in detectable urinary concentrations of WADA-prohibited cannabinoids in urine. Exercise appeared to increase concentrations of these cannabinoids. Therefore, athletes should avoid consuming broad-spectrum CBD products, given the potential associated anti-doping risks.

Detection of injury biomarkers in sweat of collegiate athletes pre- and post-football season: A pilot study. Neurotrauma Reports, Vol 6: 1-6, 2025.

Abstract - The sports medicine community and society at large have recognized traumatic brain injury (TBI) as a major public health concern. It is estimated that more than 150 million youths have played football in the United States. As an alternative to blood, sweat is a potential source for protein biomarkers, providing a non-invasive method for objective measurements for head safety guidelines. This pilot study explores sweat as a means of detecting protein biomarkers of brain injury before and

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after a football season. Participants were football players from an NCAA Division III college (*N* = 34 pre-season, *N* = 18 post-season). At pre- and post-season time points, demographic, injury history, and physical activity assessments were conducted, including application of a non-invasive sweat patch for approximately 24 h. Sweat protein biomarkers total-tau, neurofilament light, glial fibrillary acidic protein, and ubiquitin carboxyl-terminal hydrolase L1 (UCH-L1) were measured via immunoarray. Paired and un-paired non-parametric statistical analyses were conducted. Athletes reported little to no concussion injuries from the season and experienced minimal symptoms. There was a significant increase in pre-season GFAP and UCH-L1 protein levels in athletes with a history of TBI compared to those without. Comparing between pre- and post-season, there was an increase in total-tau and UCH-L1 levels. These data suggest that sweat may be a viable biofluid to assess head injury using hallmark TBI biomarkers.

Down, set, record: Assessing warm-up structure and use of neuromuscular training exercises in Canadian adolescent tackle football. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2025.

Abstract - Implementing injury prevention strategies is critical for coaches and sport organizations to promote player safety. Neuromuscular training (NMT) warm-up programs that involve aerobic, strength, agility, balance/coordination, and head-and-neck control exercises have demonstrated injury rate reductions of >35% across many adolescent sports. Tackle football (i.e., American football) has amongst the highest injury rates within youth sport and lacks examination of current warm-up structure and exercises. Our video-analysis study examined footage from 23 practice and game warm-ups (46 total) to document the duration and exercises completed for nine adolescent (males, ages 14–17) tackle football teams over a 9-week season within Calgary (Canada). We found that teams typically spent less than five minutes actively warming-up with most exercises being aerobic (51%) and dynamic stretching (29%), while <10% of exercises involved static stretching (6%) or NMT components of strength (9%), agility (3%), balance/coordination (1%), and head-and-neck control (1%). Warm-up length and use of NMT components were similar between games and practices, and over the season. Our findings demonstrate that NMT exercises are not being used in tackle football, which supports future implementation and

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evaluation of a tackle football-specific NMT program. We also provide coaches with a method for examining their team's warm-up structure.

Effects of sleep deprivation on sports performance and perceived exertion in athletes and non-athletes: A systematic review and meta-analysis. Frontiers in Physiology, 16: 1544286, 2025.

Background: Sleep deprivation can significantly affect sports performance and the perception of fatigue. However, the impact of sleep deprivation on sports performance remains a subject of ongoing debate across different populations.

Objectives: This study aimed to investigate the effects of sleep deprivation on sports performance and ratings of perceived exertion (RPE) in different groups, as well as how different types of sleep deprivation affect these aspects.

Methods: This systematic review followed the PRISMA guidelines (PROSPERO CRD42023492792). Randomized controlled trials (RCTs) and randomized crossover studies published in any language or up to any date were eligible based on the P.I.C.O.S. criteria. The systematic search included databases such as PubMed, Cochrane, Embase, Web of Science, and EBSCO, covering studies up to September 2024. The Cochrane RoB 2 tool was used to assess the risk of bias. Meta-analysis was conducted using either a fixed-effect model or a random-effects model. This study conducted subgroup analyses based on different populations, types of sleep deprivation, and testing times.

Results: This review includes 45 studies, comprising 16 on aerobic endurance (AE) performance, 8 on anaerobic endurance (AnE) performance, 23 on explosive power (EP), 10 on maximum force (MF), 4 on speed, 4 on skill control, and 12 on rating of perceived exertion (RPE). The results indicate that sleep deprivation significantly impaired AE in athletes [SMD = -0.66; 95% CI (-1.28, -0.04); $P = 0.04$], as well as EP [SMD = -0.63; 95% CI (-0.94, -0.33); $P < 0.00001$], MF [SMD = -0.35; 95% CI (-0.56, -0.14); $P = 0.001$], speed [SMD = -0.52, 95% CI (-0.83, -0.22); $P = 0.0008$], skill control [SMD = -0.87; 95% CI (-1.7, -0.04); $P = 0.04$], and RPE [SMD = 0.39; 95% CI (0.11, 0.66); $P = 0.006$]. Additionally, AE was also reduced in healthy non-athletes [SMD = -1.02; 95% CI (-1.84, -0.21); $P = 0.01$]. During the sleep deprivation process, early

sleep deprivation (PSDE) significantly reduced EP [SMD = -1.04; 95% CI (-1.58, -0.5); $P = 0.0002$], MF [SMD = -0.57; 95% CI (-0.94, -0.19); $P = 0.003$], speed [SMD = -0.78; 95% CI (-1.35, -0.2); $P = 0.008$], and RPE [SMD = 0.6; 95% CI (0.17, 1.02); $P = 0.006$]. Late sleep deprivation (PSDB) impacted speed [SMD = -0.57; 95% CI (-1.15, 0.01); $P = 0.05$], skill control [SMD = -2.12; 95% CI (-3.01, -1.24); $P < 0.00001$], and RPE [SMD = 0.47; 95% CI (0.02, 0.92); $P = 0.04$]. Overall, total sleep deprivation primarily affected AE [SMD = -0.56; 95% CI (-1.08, -0.05); $P = 0.03$]. In terms of testing phases, p.m. tests had a significant impact on AE [SMD = -1.4; 95% CI (-2.47, -0.34); $P = 0.01$], EP [SMD = -0.68; 95% CI (-1.06, -0.31); $P = 0.0004$], MF [SMD = -0.3; 95% CI (-0.51, -0.09); $P = 0.005$], skill control [SMD = -2.12; 95% CI (-3.01, -1.24); $P < 0.00001$], and RPE [SMD = 0.72; 95% CI (0.20, 1.24); $P = 0.007$]. In contrast, a.m. tests primarily affected speed [SMD = -0.81; 95% CI (-1.52, -0.1); $P = 0.03$] and RPE [SMD = 0.44; 95% CI (0.01, 0.86); $P = 0.04$].

Conclusion: Sleep deprivation significantly impairs athletes' performance across various domains, including AE, MF, speed, and skill control, while also exacerbating RPE. In contrast, although sleep deprivation also negatively affects the AE of healthy non-athletes. Furthermore, PSDE appears to have a more pronounced effect on sports performance overall. Additionally, performance assessments conducted in the p.m. have been shown to further impact sports performance. These findings are crucial for understanding how sleep deprivation impacts both athletes and non-athletes, particularly in the context of training and competitive settings.

Evening smartphone exposure impairs sleep quality and next-day performance in elite soccer players: A randomized controlled trial. Biology of Sport, 43: 227-242, 2026.

Abstract - This study aimed to examine the effects of pre-bedtime smartphone use on sleep quality and athletic performance in soccer players while also investigating potential time-of-day variations. In this randomized controlled crossover trial, 16 male elite-level players were assigned to either use a smartphone for two hours prior to bedtime or read magazines (control), separated by a one-week washout period. Participants completed morning and afternoon performance tests (cognitive and physical assessments) and sleep quality measurements. Nocturnal-smartphone use significantly

impaired sleep quality, increasing sleepiness after days 3 and 5 ($p < 0.01$; d=5.74, d=5.72, respectively), decreasing total sleep time, increasing sleep onset latency, and reducing sleep efficiency (all $p < 0.01$; d=1, d=4.59). Cognitive performance initially showed improved afternoon results, although following five days of smartphone use, this pattern reversed with enhanced morning performance ($p < 0.01$; d=0.53, d=1.48). Simple and choice reaction times deteriorated significantly in afternoon sessions compared to both baseline and control conditions ($p < 0.01$; d=0.96–3.47). Physical performance tests revealed decreased jumping ability and slower reactive agility times following five nights of smartphone use, particularly in afternoon sessions ($p < 0.01$; d=0.85 0.91). Five consecutive nights of pre-bedtime smartphone use impaired sleep quality and both cognitive and physical performance in elite soccer players, with stronger effects in afternoon sessions. These findings emphasize the importance of implementing device-free periods prior to bedtime and potentially adjusting training schedules when evening screen exposure is unavoidable. Future research should explore countermeasures for managing evening device exposure in elite athletes.

Hamstring injury mechanisms and eccentric training-induced muscle adaptations: Current insights and future directions. Sports Medicine, Ahead of Print: 1-15, 2025.

Abstract - Hamstring injuries are a major concern in sports owing to their high incidence and recurrence rates, highlighting the need for a deeper understanding of their mechanisms and prevention. This narrative review aims to inform hamstring injury prevention strategies by examining: (1) the causes of hamstring injuries, (2) the effectiveness of eccentric training in reducing injury risk, and (3) muscle adaptations from eccentric training that may offer protective effects. Hamstring injuries often occur during the late swing phase of running, potentially due to insufficient or delayed neural activation or an inability to generate the necessary force to decelerate the leg and resist active overstretching. In this phase, the hamstrings must produce large eccentric forces while operating at long lengths, placing them in a vulnerable position. Despite the potential of eccentric training to induce muscle adaptations that may reduce injury risk, current research has overly focused on architectural changes, particularly resting fascicle lengthening, without adequately exploring how these adaptations influence the functional behavior of hamstrings during exercise. In addition, the lack of research into adaptations of non-

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contractile and neural elements in the hamstrings following eccentric training represents a significant gap in the literature. This review argues for a broader focus on these underexplored areas to enhance hamstring injury prevention strategies. Further research is essential to fully understand the mechanisms behind muscle fascicle lengthening after eccentric training. Exploring functional and regional differences in hamstring adaptations and delving deeper into non-contractile and neural elements could enhance injury prevention strategies, potentially reducing the incidence of hamstring injuries.

Heavy hitters, light sleepers: Collision frequency and locomotor load on sleep architecture in professional Rugby Union players. European Journal of Sport Science, 25: e70052, 2025.

Abstract - To assess whether certain players are more vulnerable to post-match sleep disturbances by examining the relationship between match demands-collision frequency and locomotor load-and sleep in professional male rugby union players. A linear mixed-effects regression examined the relationship between match variables and sleep in 13 rugby players across three matches. Match variables included six physical demand variables derived from video analysis and GPS data (collision frequency, total distance, high-speed distance, sprint distance, acceleration load and fast acceleration count) and two contextual variables (location and kick-off time). Sleep variables collected via home-based polysomnography included total sleep time, sleep efficiency, sleep onset/offset, sleep onset latency, wake after sleep onset, number of awakenings and sleep stages (light, deep and rapid eye movement sleep [REM], evaluated both by proportion [%] and time [min]). Each match collision decreased total sleep time ($\beta = -4 \pm 1$ min and $p = 0.006$) and REM sleep (time: $\beta = -2 \pm 0$ min and $p < 0.001$; proportion: $\beta = -0.6 \pm 0.2\%$ and $p = 0.021$). Conversely, every 500 m increase in locomotor load (total distance) increased REM sleep (time: $\beta = +6 \pm 2$ min and $p = 0.014$; proportion: $\beta = +2.7 \pm 0.6\%$ and $p = 0.002$). Every 100 m increase in high-speed distance was associated with decreased REM sleep time ($\beta = -7 \pm 3$ min and $p = 0.020$). Match demands, including collision frequency and locomotor load, were associated with changes in post-match sleep architecture in professional rugby players, particularly REM sleep. Furthermore, greater number of collisions was associated with reduced sleep quantity. Practitioners can leverage GPS and video analysis data to tailor additional sleep strategies aimed at improving post- match sleep based on individual match demands.

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Hip internal rotation restriction and heel raise mechanics as risk factors for Jones fractures in youth football players. Asia-Pacific Journal of Sports Medicine, Arthroscopy, Rehabilitation and Technology, 42: 15-20, 2025.

Background: Jones fractures, stress fractures at the proximal metaphyseal-diaphyseal junction of the fifth metatarsal, are common in sports that involve repetitive jumping and turning, particularly football (i.e., soccer). The incidence among Japanese football players is higher than in Europe, with a high refracture rate, particularly in conservatively managed cases. Although surgical treatment is preferred for athletes, limited data exist on risk factors in youth players. This study aimed to investigate the prevalence and risk factors for Jones fractures among high-level Japanese high school football players through medical examinations.

Methods: Medical examinations were conducted during the 2024 tournament involving 20 top-level high school teams (567 players). Ultrasound and X-ray imaging identified fractures, and players underwent various physical tests, including hip internal rotation (HIR) and weight-bearing position during a single heel raise (SHR). A questionnaire collected demographic data and fracture histories. Statistical analyses, including the Wilcoxon signed-rank test and Mann-Whitney *U* test, were conducted to determine associations with fracture risk.

Results: Of 232 male players (40.9 % of all), 18 (7.8 %) tested positive on ultrasound, seven (3.0 %) had confirmed fractures on radiography, and 11 reported a history of Jones fractures. Players with restricted HIR <30° and lateral weight-bearing during SHR showed a significantly higher fracture risk (Crude Odds Ratio: 3.74, $p = 0.01$). Additional lateral weight bearing during SHR increased the risk (Crude Odds Ratio, 4.02; $p = 0.007$).

Conclusion: Restricted HIR and lateral weight-bearing during SHR were identified as risk factors for Jones fractures in youth football players. Recognizing these factors may enable targeted preventive measures to reduce injury risk in high-level athletes.

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Initiating testosterone therapy without indication for treatment of low testosterone is associated with higher risk of tendon tear. Sports Health, Ahead of Print: 1-7, 2025.

Background: Studies have demonstrated an increasing rate of testosterone supplementation in patients who do not meet criteria for testosterone deficiency, but the risks associated with testosterone supplementation are unclear.

Hypothesis: Patients initiating testosterone therapy would be diagnosed with significantly more tendon tears and undergo significantly more tendon repairs than matched controls never on testosterone therapy.

Study design: Cohort study.

Level of evidence: Level 3.

Methods: Male patients who initiated testosterone therapy within 30 days of a primary care visit and who did not meet criteria for treatment of low testosterone were identified (n = 112,242). Propensity score matching generated a control cohort (n = 448,968) with no history of testosterone therapy and based on several comorbidities at the time of said age-matched primary care visit. Rates of tendon tears and tendon repairs within 1 and 2 years after primary care visit were compared using multivariable logistic regression.

Results: Within 1 and 2 years, compared with controls, the testosterone supplementation cohort demonstrated significantly higher rates of rotator cuff tear (odds ratio [OR], 1.35; OR, 1.31), patellar tendon tear (OR, 2.06; OR, 2.14), and any tendon tear (OR, 1.33; OR, 1.31). The testosterone supplementation cohort demonstrated significantly higher rates of quadriceps tendon tears within 1 year (OR, 1.59). Within 1 and 2 years, compared with controls, the testosterone supplementation cohort demonstrated significantly higher rates of rotator cuff repair (OR, 1.28; OR, 1.23), distal biceps tendon repair (OR, 1.65; OR, 1.54), patellar tendon repair (OR, 2.28; OR, 2.18), and any tendon repair (OR, 1.27; OR, 1.23).

Conclusion: Testosterone supplementation is associated with an increased risk of tendon tears and repairs for at least the first and second year after initiating therapy in patients with no documented

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laboratory testing. Patients considering testosterone therapy should be counseled on this increased risk of tendon injury.

Knee injury mechanism varies by injury category: Video analysis of multi-ligament knee injuries in the National Football League. Knee Surgery, Sports Traumatology, Arthroscopy, Ahead of Print: 1-8, 2025.

Purpose: This study aims to characterize the mechanism of multi-ligament knee injury (MLKI) sustained during a National Football League (NFL) game through video analysis.

Methods: A retrospective video analysis of official NFL game footage spanning 1997-2022 was performed. Players with MLKIs were identified from publicly available injury surveillance data. Player and game demographic information was collected. Athletes were categorized based on ligament injury patterns. Videos including MLKI were analyzed to categorize injuries based on non-contact, indirect or direct contact mechanisms and determine the hip, knee and ankle position at the time of injury. Differences across MLKI groups were identified by Fisher's exact test.

Results: Thirty-five MLKIs were identified. MLKIs most commonly occurred from direct contact to the limb (80%), with the second most common injury mechanism being indirect contact (14.3%). The most common MLKI was injury to the anterior cruciate ligament (ACL) and medial collateral ligament (MCL) (65%), while the second most common was the ACL and lateral collateral ligament (LCL) (15%). Direct contact MLKI most often resulted in a valgus and external rotational force about a flexed knee (58%). There were statistically significant differences when comparing the direction of contact, hip and knee position between injury MLKI groups. Combined ACL and MCL injury commonly occurs with knee flexion, valgus and external rotation. Combined ACL/posterior cruciate ligament (PCL) or ACL/LCL injuries occurred most commonly with varus and internal rotational forces on an extended knee. Multiligament injuries involving three ligaments occurred only from direct contact mechanisms.

Conclusion: Various injury mechanisms and characteristics result in different types of MLKI in NFL athletes, though the vast majority of MLKIs occur from direct contact forces to the knee. Combined ACL

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and MCL injury occurred most frequently, resembling the mechanism and lower extremity position for isolated ACL injury, though it frequently occurs due to direct contact with the knee.

Level of evidence: Level IV.

Longitudinal changes in strength, power, speed, and body mass in American football players: Diminishing returns as training age increases. International Journal of Sports Science & Coaching, Ahead of Print: 1-13, 2025.

Abstract - The aims of this study were to: 1) detail the strength, power, speed, and body mass (i.e. physical qualities) of National Collegiate Athletic Association (NCAA) Division 1 American football players by playing position across a four-year collegiate career, and 2) quantify the rate of change in physical qualities as athletes progress through their eligibility period. 2628 observations from 512 NCAA Division 1 American football players were collected during a standardized testing battery that took place across a 15-year period. One-repetition maximum bench press and back squat, 40-yard sprint, vertical jump, standing broad jump, and body mass were analyzed. Year-on-year changes in physical characteristics by the entire cohort and positional groups were analyzed using linear mixed models with Cohen's effect size (ES) ± 95% confidence intervals (CI). Improvements in physical qualities occurred over the four years, although the largest changes were evident in the first year, with the improvements becoming less pronounced each year. For example, across the cohort, *large* effects in the bench press (ES ±95%CI: 1.63 ± 0.18) and back squat (ES ±95%CI: 1.62 ± 0.18) occurred in the first year but only *small* changes occurred between years 3 and 4 (ES ±95%CI: 0.45 ± 0.16 and 0.26 ± 0.16, respectively). Collectively, this study demonstrates the longitudinal changes in strength, power, speed, and body mass of collegiate Division 1 American football players across different positional groups. Furthermore, it shows that physical qualities improve throughout the four years, but the year-on-year change becomes smaller each year and therefore suggests physical qualities become harder to increase/improve.

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Neurosciences and Sports Rehabilitation in ACLR: A narrative review on winning alliance strategies and connecting the dots. Journal of Functional Morphology and Kinesiology, 10: 119, 2025.

Abstract - This narrative review explores the significant evolution of sports rehabilitation, tracing its trajectory from basic exercise therapies of the early 20th century to the advanced, neuroplasticity-driven approaches of the 21st century, with a specific focus on anterior cruciate ligament reconstruction (ACLR). The primary aim is to understand how neuroplasticity, motor control, and sensorimotor retraining can optimize recovery, reduce reinjury risk, and enhance long-term athletic performance, and to synthesize current rehabilitation strategies that integrate innovative technologies, such as robotics, virtual reality (VR), and biofeedback systems, to address the neurocognitive deficits that contribute to the alarmingly high reinjury rates (9-29%) observed in young athletes post-ACLR. These deficits include impaired proprioception, motor control, and psychological factors like fear of reinjury. The methodology employed involves a narrative review of peer-reviewed literature from databases including PubMed, Scopus, and Web of Science. The synthesis of findings underscores the importance of holistic rehabilitation approaches, including targeted proprioceptive exercises, dual-task drills, and immersive VR training, in enhancing sensorimotor integration, decision-making, and athlete confidence. Furthermore, this review highlights the critical need for long-term monitoring and interdisciplinary collaboration between neuroscientists, physiotherapists, and engineers to refine rehabilitation protocols and ensure sustained recovery. By leveraging neuroplasticity and advanced technologies, the field can shift from a focus on purely physical restoration to comprehensive recovery models that significantly reduce reinjury risks and optimize athletic performance.

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Periodization for Success – In-season external training loads relative to competition load in American football. Frontiers in Sports and Active Living, 7: 1662240, 2025.

Introduction: Despite an exponential development in performance monitoring technologies, the physical performance demands of sport remain an understudied topic in scientific literature. Thus, the primary purpose of this study was to quantify and compare the training loads (TL) of a collegiate American football team between in-season practices and official games by general position group, event type, and to assess the interaction between the two.

Methods: Twenty-seven NCAA Division-II athletes volunteered to participate in this investigation. In-season TL during 32 practices (categorized as days before game day; GD minus) and 11 conference games were recorded using global positioning system technology. Collected data included total duration, total distance, yards traveled per minute, hard running distance, hard running efforts, 2-dimensional (2D) load, and 3-dimensional (3D) load.

Results: A factorial analysis of variance revealed significant main effects in TL for event type ($p < 0.001$) and position groups ($p < 0.001$), and an interaction effect between the two ($p < 0.001$). Unique microcyclic characteristics were observed for each measure of interest. Relative to game values (100%), values for training duration (+25% to −12%; GD-4 to GD-1), yards per minute (+15% to −11%), total distance (+37% to −3%), hard running distance (+33% to −7%), hard running efforts (+33% to −12%), 2D-load (+40% to −7%), and 3D-load (+44% to −3%) were significantly greater than game values on distinct days during the week.

Discussion: These findings can improve the current understanding of practice demands relative to games, which may support more optimal sport-specific periodization approaches within American football.

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Pharmacological and non-pharmacological strategies for enhancing sleep quality in athletes: Implications for performance and recovery. Quality in Sport, 41: 60176, 2025.

Abstract - Sleep is a critical determinant of recovery, performance, and overall well-being in athletes. Despite its importance, sleep disturbances—including insomnia, delayed sleep onset, and circadian rhythm disruption—are common in athletic populations, driven by both primary factors (e.g., overtraining, evening workouts, jet lag) and secondary factors (e.g., stress, injury, metabolic conditions). These disturbances can impair physical restoration, hormonal balance, and cognitive functioning. This review explores the physiological foundations of sleep in athletes, outlines the multifactorial causes of insomnia, and evaluates pharmacological strategies—including melatonin, non-benzodiazepine hypnotics, cannabidiol (CBD), and certain antidepressants—for managing sleep disorders. While pharmacotherapy may provide short-term benefits in select situations, non-pharmacological approaches such as Cognitive Behavioral Therapy for Insomnia (CBT-I) remain the gold standard due to their long-term efficacy and minimal side effects. An integrated, personalized approach that combines behavioral, physiological, and pharmacological strategies offers the greatest potential for optimizing sleep and supporting sustainable athletic performance.

Physicians' knowledge, attitudes, and perceptions about medical cannabis in the United States: A scoping review. Medical Cannabis and Cannabinoids, 8: 58-64, 2025.

Introduction: As legalization of medical cannabis (MC) in the USA expands, there remains uncertainty in clinical guidance. Healthcare professionals remain unprepared to communicate to patients the therapeutic outcomes and possible adverse effects of MC utilization. There is limited training provided at all levels of medical education, even for professionals with many years of clinical practice. Additionally, there is minimal scientific research, which delays the development of evidence-based guidelines.

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Methods: This review followed established methodological approaches for scoping reviews according to PRISMA-ScR guidelines. Studies were included if they addressed the attitudes and beliefs of medical practitioners in the USA and were published after the year 2000.

Results: There were forty-one studies from January 2013 to February 2025 included in the format of both electronic surveys and qualitative interviews. Participants included US physicians, other healthcare professionals, and medical trainees, representing multiple clinical specialties. Physicians reported lack of confidence in counseling patients or managing their use of MC. Oncologists, emergency medicine physicians, pain management specialists, and primary care physicians perceived that MC is beneficial for managing chronic pain, nausea, loss of appetite, depression, and other symptoms. Obstetric providers had unfavorable perceptions about perinatal use of MC. Physicians practicing in states where the drug has been legalized and those with greater years of practice were more comfortable recommending MC and counseling patients.

Conclusions: US physicians and medical trainees perceived significant knowledge barriers to recommending MC and counseling patients on its therapeutic use. Implementing clear clinical practice guidelines, further education on these drugs in clinical curriculums, and enhancing continuing education offerings would improve prescriber confidence. Increased research could also assist medical professionals in appropriate clinical decision making.

Preferences including graft choice, lateral augmentation, and rehabilitation after anterior cruciate ligament reconstruction among National Football League team orthopaedic surgeons. Arthroscopy, Sports Medicine, and Rehabilitation, Vol.7(4): 1-8, 2025.

Purpose: To assess current treatment preferences among National Football League (NFL) team orthopaedic knee surgeons for the management of anterior cruciate ligament (ACL) injuries.

Methods: NFL team physicians who are sports medicine knee surgeons were identified by using public databases and emailed a multiple-choice poll evaluating various preferences involving treating ACL injuries in various situations.

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Results: Ninety-seven NFL team knee surgeons were identified, with 51 responding (53%). Respondent surgeons had 19.6 years of experience performing ACL surgery with 103.9 ACL reconstructions performed annually. Patella tendon was the favored graft in a 20-year-old running back, as well as the 25-year-old and 35-year-old recreational athlete. Quadriceps tendon became the second most favored graft in all situations. In total, 4% of surgeons said they would perform a primary repair of an ACL in the setting of a proximal tear in a 20-year-old running back, 96% said they would consider adding a lateral augmentation procedure in the primary ACLR setting but only if certain findings were obtained, 25% recommended outpatient physical therapy to supplement working with the team rehabilitation staff, whereas 76% did not recommend a brace for return to play.

Conclusions: Patella tendon autograft is the favored graft in ACL reconstruction among professional and recreational athletes by NFL team orthopaedic knee surgeons. Quadriceps tendon autograft is the second most preferred graft. Lateral augmentation procedures would be considered in a primary ACLR but only if certain findings are present. Most respondents do not recommend supplementing rehab with outpatient PT or using a brace for return to play.

Clinical relevance: NFL team surgeons consider numerous factors when treating ACL injuries in professional athletes, including return to function, speed of recovery, and risk of reinjury, all of which directly impact the athlete's ability to sustain a livelihood during a relatively short professional career. Insights gained from these surgical practices may inform and enhance treatment strategies across a broader patient population.

Reframing Recovery: Fantasy sports metrics as scalable tools for measuring post-injury athletic performance. Cureus, 17(9): e91602, 2025.

Abstract - While return to play has traditionally been the benchmark for evaluating athlete recovery after injury, it does not adequately capture whether athletes regain their prior level of performance. The concept of return to pre-injury performance (RPIP) offers a more meaningful measure but lacks a standardized approach for quantification across sports. This narrative review examined the literature using fantasy sports scores and wins above replacement (WAR) to evaluate post-injury performance among professional athletes and to assess their utility as foundations for RPIP. A targeted PubMed

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search identified studies that used fantasy scores or WAR to quantify post-injury performance in the National Football League (NFL), National Basketball Association (NBA), National Hockey League (NHL), and Major League Basketball (MLB), requiring stratification by injury and use of publicly available scoring systems. Six studies encompassing 15 sports-pecific cohorts and 785 athletes met the inclusion criteria. Fantasy metrics and WAR consistently reflected post-injury performance declines in 14 of 15 cohorts across different sports and injury types, despite variation in study design, scoring systems, and follow-up windows. However, only two studies explicitly referenced RPIP. Fantasy scores and WAR may serve as scalable tools for quantifying post-injury performance and provide a preliminary framework for standardizing RPIP across sports. Their public availability, longitudinal structure, and comprehensiveness make them uniquely suited for research, clinical, and operational use. Formalizing RPIP around these metrics could significantly enhance recovery evaluation and decision-making in sports medicine.

Resident Case Series: The utility of the Athletic Should Test using wireless portable force plates for measuring peak force in NCAA Division 1 collegiate quarterbacks. International Journal of Sports Physical Therapy, 20(7): 1039-1049, 2025.

Background: Given the high incidence of shoulder injuries among quarterbacks due to the repetitive stress of the overhead throwing motion, there is a need for targeted, sport-specific assessments that reflect the functional demands of throwing. The Athletic Shoulder (ASH) test, previously validated in rugby and baseball populations, offers a potential solution for upper extremity strength assessment in football quarterbacks. The primary purpose of this case series was to explore the feasibility of conducting the ASH test as a method for evaluating upper extremity strength in overhead athletes, specifically collegiate quarterbacks.

Study design: Case series.

Methods: Three NCAA Division I quarterbacks participated in weekly ASH testing over a nine-week period. Peak isometric force, peak force normalized to body weight, and limb symmetry indices were

measured in the shoulder "Y" and "T" positions using portable force plates. Testing was completed pre-practice on non-game days to minimize fatigue effects and maintain consistency.

Results: The dominant shoulder consistently produced greater peak force than the non-dominant side in both Y and T positions. The mean dominant-to-non-dominant limb symmetry index was 1.13 in the Y position and 1.14 in the T position. Within limbs, peak force was greater in the Y position than in the T position. In both positions and in both arms, athletes generated at least 14% of their body weight in force on average.

Conclusion: These findings can provide insight into strength asymmetries and functional performance benchmarks, enabling strength coaches, athletic trainers, and rehabilitation professionals to fine-tune training and rehabilitation programs. Incorporating the ASH test into a standardized assessment battery may enhance the ability to evaluate an athlete's readiness to train or compete, promoting a proactive approach to performance optimization and injury prevention. Its utility and actionable metrics make the ASH test a practical tool for in-season monitoring, allowing practitioners to make informed, data-driven adjustments throughout the season.

Level of evidence: Level 4.

Return to play and performance after patellar fracture in American professional sports: A case-control cohort analysis. The Physician and Sportsmedicine, Ahead of Print: 1-12, 2025.

Objectives: This study describes the effects of patellar fracture on return to play (RTP) and functional outcomes among athletes in American professional sports.

Methods: Professional athletes from the National Football League (NFL), National Basketball Association (NBA), Major League Baseball (MLB), and National Hockey League (NHL) who suffered a patellar fracture between January 1965 and December 2021 were identified through injury reports and public archives. Performance scores, play time, and games played were collected for the season preceding patellar fracture and 2 seasons after RTP, and differences in recorded metrics compared to pre-injury levels and matched controls were analyzed.

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Results: Twenty-nine of 41 (71%) injured athletes returned to play at an average of 217 days. Among all players, play volume decreased in year 1 compared to baseline but recovered to pre-injury levels in year 2. Athletes treated operatively experienced an initial decline in performance ($p < 0.01$) but recovered to pre-injury performance level in year 2. Nonoperative management resulted in a decline in performance in year 2 of RTP ($p = 0.02$). Athletes treated operatively performed significantly worse than matched controls in year 1 of RTP (64% vs. 99%; $p = 0.04$) but recovered to a similar level of performance as controls in year 2 (87% vs. 91%; $p = 0.90$).

Conclusion: A 71% rate of RTP was demonstrated among this limited cohort of 29 athletes in American professional sports after isolated patellar fracture. Although details regarding fracture characteristics and operative reports were not available for analysis, operative management was not associated with longer absence from play compared to nonoperative treatment. Despite the limitations of this study, the findings suggesting operative management may improve prospects of maintaining elite performance following RTP warrant further investigation.

Level of evidence: Case-control cohort analysis; Level of evidence, 3.

Sleep architecture immediately after a sport-related concussion sustained during a professional Rugby Union match: A retrospective case study. International Journal of Sports Physiology and Performance, Ahead of Print: 1-6, 2025.

Purpose: A sport-related concussion (SRC) can cause sleep disturbances within 72 hours of injury. However, most research focuses on sleep after >72 hours, with limited attention to acute sleep responses, despite their potential impact on SRC recovery.

Methods: This case study examines a 22-year-old professional rugby union player who sustained an SRC during an away match. Sleep architecture (sleep stages: N1, N2, N3, and rapid eye movement [REM] sleep) and physiological responses during sleep over 4 nights were analyzed-sleep before injury (PRE), the 2 nights immediately following injury (INJ and INJ + 1), and after recovery from injury (POST)-using home-based polysomnography. Return transmeridian travel was undertaken on INJ + 1.

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Results: The player took 15 minutes to fall asleep and slept <6 hours on INJ, with reductions in N3 (19% vs 22%-37%) and REM sleep (18% vs 20%-32%) compared with all other nights. Despite increased N3 and REM sleep on INJ + 1, sleep disturbances persisted (23 min to fall asleep, 69 min of wake after sleep onset, 40 awakenings, and sleep efficiency below 85%). Compared with sleep on PRE and POST, mean oxygen saturation and heart-rate variability were lower, mean heart rate was elevated, and respiratory events during sleep were greater on INJ and INJ + 1.

Conclusion: Acute responses after the SRC included sleep disturbances and elevated sympathetic activity (during sleep). The player had difficulty achieving deep, restorative sleep the first night following the SRC, aligning with the non-restorative-sleep complaints commonly reported by concussed athletes on the night(s) immediately following injury.

Smooth pursuit velocity after a season of repetitive head impacts in American football players. JAMA Ophthalomology, 18: e252935, 2025.

Importance: Repetitive head impacts (RHI) are blows to the head that do not elicit clinical signs or concussion symptoms. Yet, research suggests that brain integrity and functionality can be altered following a single season of contact collision sports. These deficits are untraceable by current clinical testing.

Objective: To examine the association between RHI and oculomotor control across a single division I American football season.

Design, setting, and participants: This single-center cohort study took place from July 2020 to May 2022. Data were analyzed from May 2022 to December 2024. Participants included 25 division I football players (11 high-dose and 14 low-dose; mean age, 20 [SD, 3] years) and 10 controls (mean age, 22 [SD, 1] years). High-dose and low-dose groups were determined by the head impacts incurred during the season using instrumented mouthguards.

Exposure: A single competitive season of contact sports.

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Main outcomes/measures: All football players wore instrumented mouthguards throughout the 2 seasons (2021 to 2022). Each season was treated as an independent observation with no individual football players included in more than 1 season. A random sample of head impacts across the season were video verified. All participants completed an eye tracking task that consisted of following a Landolt-C moving horizontally during fast (90° per second) and slow (30° per second) conditions at preseason (PRE), midseason (MID), and postseason (POST). Smooth pursuit eye movement velocity was calculated and analyzed.

Results: A total of 25 male division I American football student-athletes (RHI group; mean age, 20 [SD, 3] years; mean height, 185.72 [SD, 7.40] cm; mean weight, 104.02 [SD, 15.69] kg) and 10 healthy noncontact controls (7 recreationally active [4 female, 3 males] and 3 noncontact National Collegiate Athletic Association swimmers [all female]; mean age, 22 [SD, 1] years) participated in this cohort study. During the season, smooth pursuit eye movement velocity was PRE, 10.01 (SD, 3.16)° per second; MID, 11.48 (SD, 6.57)° per second; and POST, 11.10 (SD, 8.65)° per second for the high-dose group, while the low-dose group was PRE, 11.28 (SD, 4.10)° per second; MID, 11.50 (SD, 5.86)° per second; and POST, 15.02 (SD, 8.26)° per second, with the controls being PRE, 15.60 (SD, 4.44)° per second; MID, 17.95 (SD, 3.73)° per second, and POST, 13.44 (SD, 8.54)° per second. Specifically, at PRE, RHI high-dose players had slower smooth pursuit eye movement velocity compared with controls (difference, 4.28; 95% CI, 2.68-5.81; P = .01). At MID, both high-dose (difference, 8.49; 95% CI, 5.18-11.81; P = .01) and low-dose (difference, 9.15; 95% CI, 6.42-11.87; P = .02) groups had slower smooth pursuit eye movement velocities than controls.

Conclusions and relevance: In this study, smooth pursuit eye movement velocity was not affected during a single season of contact sports. Group differences suggest existing deficits before the season begins, possibly due to prior contact sport history. Interpretations should be tempered by the small sample size, single institution used for data collection, and the exploratory nature of the study.

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Tackle characteristics associated with concussion in elite men's rugby union: Unpicking the differences between tacklers and ball-carriers. BMJ Open Sport & Exercise Medicine, 11: e002612, 2025.

Objective: To identify characteristics of tackling, of being tackled and interactions between tackle characteristics that are associated with concussion.

Methods: A case-control study in male professional rugby union players in England over five seasons (2016/2017 to 2020/2021) analysed characteristics of tackles that led to a clinically diagnosed concussion (cases), and a control group of tackles that did not result in a concussion. ORs were plotted against the overall frequency of each tackle characteristic.

Results: 231 tackles resulting in concussions (tackler 178, 77%; ball-carrier 53, 23%), alongside 9963 control tackles, were analysed. For tacklers, 'head to torso' ($_{\text{Lower CI}}$ OR $_{\text{Upper CI; 4.0}}$ 6.5 $_{10.7}$) had relatively low odds of concussion compared with 'no head contact'. 'Head to knee' had the highest odds of concussion ($_{75.3}$ 155.8 $_{322.4}$), but 'head to hip' occurred more frequently and had the highest number of concussions per 1000 tackles (3.1/1000 tackles). For ball-carriers, 'head to head' contact had the highest odds of concussion ($_{56.7}$ 104.3 $_{192.0}$). When 'tackler body position' was 'upright', the odds of concussion to the tackler were significantly higher when contacting the ball-carrier's 'head and neck' versus their 'torso' ($_{3.0}$ 23.7 $_{206.7}$).

Conclusions: Lower tackles reduce the chances of concussion to ball-carriers. The influence of tackle height on concussion to tacklers is more nuanced, but the chances are relatively low when contact is made with the ball-carrier's torso. These findings support ongoing implementation of strategies to reduce concussion risk by lowering tackle height to target the torso.

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Using a Bayesian network to classify time to return to sport based on football injury epidemiological data. PLOS ONE, 20(3): e0314184, 2025.

Abstract - The return-to-sport (RTS) process is multifaceted and complex, as multiple variables may interact and influence the time to RTS. These variables include intrinsic factors related the player, such as anthropometrics and playing position, or extrinsic factors, such as competitive pressure. Providing an individualized estimation of time to return to play is often challenging, and clinical decision support tools are not common in sports medicine. This study uses epidemiological data to demonstrate a Bayesian Network (BN). We applied a BN that integrated clinical, non-clinical factors, and expert knowledge to classify time day to RTS and injury severity (minimal, mild, moderate and severe) for individual players. Retrospective injury data of 3374 player seasons and 6143 time-loss injuries from seven seasons of the professional German football league (Bundesliga, 2014/2015 through 2020/2021) were collected from public databases and media resources. A total of twelve variables from three categories (player's characteristics and anthropometrics, match information and injury information) were included. The response variables were 1) days to RTS (1-3, 4-7, 8-14, 15-28, 29-60, > 60, and 2) injury severity (minimal, mild, moderate, and severe). The sensitivity of the model for days to RTS was 0.24-0.97, while for severity categories it was 0.73-1.00. The user's accuracy of the model for days to RTS was 0.52-0.83, while for severity categories, it was 0.67-1.00. The BN can help to integrate different data types to model the probability of an outcome, such as days to return to sport. In our study, the BN may support coaches and players in 1) predicting days to RTS given an injury, 2) team planning via assessment of scenarios based on players' characteristics and injury risk, and 3) understanding the relationships between injury risk factors and RTS. This study demonstrates the how a Bayesian network may aid clinical decision making for RTS.

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